Quantum Adapters for Frozen World Models: A Pre-Registered Study of Input-Conditioned Rydberg Bottlenecks on Public Physics Benchmarks

Anonymous Author(s)
Affiliation
Address
email

Abstract

Two claims recur in the recent literature: that analog quantum hardware is a natural substrate for simulating physical dynamics, and that small quantum modules can improve large pretrained (multi)modal models. We test both in a single world-model setting, under rules designed to make failure as visible as success. Because no practical application fits on today's qubit counts whole, our design premise is division of labor: steer a small reconfigurable register to the problem's hardest analog subprocess and let a frozen world model carry the rest. Our system inserts a $\sim$500-parameter, input-conditioned quantum bottleneck — per-qubit detunings and trainable 2D atom geometry driving Rydberg-Hamiltonian evolution, so the analog block *adapts to the simulation state* — into (i) a compact neural surrogate for PDE rollout and (ii) a frozen 14B open-weights video world model, at 30 simulated qubits (three blockade-decoupled 10-qubit registers). Evaluation is pre-registered on public benchmarks with published state of the art: our primary goal is one-step test VRMSE below the strongest official baseline on The Well's `turbulent_radiative_layer_2D` (CNextU-net, 0.1956) with paired per-sample significance, alongside long-horizon rollout stability, where every published model, including a 1.3B foundation model, degrades beyond the spatial-mean floor. Controls are parameter- and tuning-matched, with explicit kill rules and a dequantization suite; at $\leq$30 simulated qubits the block is classically simulable, so all claims are about quantum-structured inductive bias, not quantum advantage. In time-boxed pilot sprints on a single 126 GB workstation, quantum bottlenecks were the only variants to beat the persistence floor on 8-step rollout, cutting rollout error by 23–37% over a matched classical trunk while a parameter-matched MLP control *hurt*; in a three-seed replication the 30-qubit Rydberg block achieved the best mean and lowest variance (0.955 ± 0.102 vs 1.111 ± 0.174 baseline), with the variant ordering analog $>$ digital $>$ none $>$ MLP. These pilot effects are directional at current power; the full-protocol verdict — positive or negative — fine-tunes the official checkpoint (which we reproduce at 0.1934 test VRMSE) and is reported at the workshop.

1 Introduction

World models — learned simulators that roll a latent state forward under dynamics — increasingly serve as the interface between machine learning and physics, from PDE surrogacy to video generation [Ohana et al., 2024, McCabe et al., 2025, Gao et al., 2026]. Their central unsolved failure is

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physical consistency over long horizons: on The Well benchmark, the best official baseline (CNextU-net) reaches one-step test VRMSE 0.1956 on `turbulent_radiative_layer_2D` but degrades to 1.01 by rollout window 13:30 — worse than predicting the spatial mean — and even the 1.3B cross-domain foundation model Walrus exceeds this floor at long horizon [Ohana et al., 2024, McCabe et al., 2025]. Dedicated video-physics benchmarks report the same gap for generative world models [Motamed et al., 2025, Li et al., 2025].

Meanwhile, two literatures make claims that, if true, bear directly on this failure. First, *analog quantum simulation*: neutral-atom platforms natively evolve Rydberg Hamiltonians, and reservoir-computing experiments on QuEra Aquila encode inputs directly into detunings, positions, and drives of up to 108 qubits [Kornjača et al., 2024]; Hamiltonian input encoding is provably expressive, offering infinite-order nonlinearity in the encoded variables [Xia et al., 2026]. Second, *quantum modules for foundation models*: unitary and quantum(-inspired) adapters now fine-tune frozen LLMs and vision–language models at extreme parameter efficiency [Koike-Akino et al., 2025, Raj and Coyle, 2025, Aizpurua et al., 2026, Zhao et al., 2026]. Neither literature, however, has produced a matched-control win on a public physical-simulation benchmark: the closest attempt, a 5-qubit mixer in a 3D FNO surrogate, beat a larger classical model but *lost to its own parameter-matched control* [Papierz et al., 2026], and reservoir studies show apparent quantum edges evaporating under equal-tuning classical baselines [Pandey, 2026]. No quantum-hybrid entry exists on The Well, PDEBench, or LagrangeBench.

Our premise connects the two claims. Analog quantum hardware may be the natural substrate for physical dynamics, but no practical application fits on today's qubit counts whole — and real simulations are rarely uniformly quantum-hard: they mix classically easy, effectively discrete structure with a harder analog core. We therefore treat the qubit budget as a scarce resource to be *allocated*: a reconfigurable, input-parameterized register is steered toward the highest-complexity analog subprocess of the problem at hand, while the frozen world model carries the rest. If the division of labor is right, small fixed budgets become useful on current hardware, and fidelity versus qubit count becomes a measurable scaling question rather than a wait for million-qubit machines. This is the hypothesis our protocol is built to confirm or kill.

We take both claims seriously by making them falsifiable in one setting. Our contributions, ordered by novelty:

1. **The first quantum adapter inside a frozen video world model.** A quantum bottleneck at the KV-cache chunk boundary of a frozen 14B open-weights interactive world model [Gao et al., 2026, Team Wan et al., 2025], scored by official public judges [Li et al., 2025, Motamed et al., 2025] — to our knowledge the first quantum module in any video world model and the first quantum entries on either benchmark. (The similarly-named QSWM [Jiang et al., 2026] is quantum-*inspired* — complex-valued, density-matrix-like latents evaluated on cellular automata — and contains no quantum module.)

2. **Checkpoint surgery against a published number.** We insert the bottleneck at the neck of the *official released* CNextU-net checkpoint from The Well and fine-tune, targeting its published 0.1956 one-step test VRMSE [Ohana et al., 2024] with paired per-window Wilcoxon significance over the 873 official test windows. Quantum U-Net bottlenecks exist from scratch [Jain and Kalev, 2026, Hossain et al., 2026, Marcandelli et al., 2025]; no quantum work has touched The Well or any released scientific-ML checkpoint — the methodology (official-checkpoint surgery + published-number target + paired statistics) and its result are the contribution, not the architecture.

3. **An input-conditioned analog ansatz, as a composite.** The Rydberg generator is parameterized by the latent state — data-dependent detunings, gradient-trained 2D atom geometry, global drive — via exact statevector autodiff inside the classical network, and the trained object is a valid neutral-atom program (Pulser-exportable). Geometry-as-parameter is the natively realizable knob of this platform: optical tweezers assemble atoms into essentially arbitrary, reconfigurable 2D/3D arrangements [Barredo et al., 2018], now demonstrated at the 6,100-qubit scale [Manetsch et al., 2025], so a trained register layout is a deployable program, not an abstraction. Each ingredient exists in isolation [DiBrita et al., 2024, 2025, Abramavicius et al., 2025]; the composite, embedded as a layer of a predictive world model, does not. Thirty simulated qubits are realized as three blockade-decoupled 10-qubit regis-

ters — an implementation detail with prior art in patched quantum generators [Huang et al., 2021].

4. **A pre-registered, kill-rule protocol.** Qubit-scaling claims already exist in looser forms — IonQ reports monotone scaling against parameter-matched classical baselines on a pre-trained model [Kim et al., 2025], and analog reservoirs scale to 108 qubits [Kornjača et al., 2024] — so we do not claim "more qubits helps" as novel. We contribute its first *pre-registered, tuning-and-parameter-matched, significance-tested* instantiation on a public PDE benchmark: matched MLP/orthogonal/tensor-network/product-state controls [Pandey, 2026, Bowles et al., 2024], explicit kill rules (if controls match, the negative result is the paper), and a dequantization suite (bond-dimension sweeps, product-state controls, finite-shot gradients), under the honest-framing standard of Schuld and Killoran [2022] and the classical-simulability caveat of Cerezo et al. [2025].

2 Problem definition

Task. Let $x_t \in \mathbb{R}^{C\times H\times W}$ be a physical field snapshot (here: density, pressure, velocity on a 128×384 grid). Given a history $(x_{t-3},\dots,x_t)$, predict x_{t+1}; long-horizon behavior is evaluated by autoregressive rollout. The official metric is variance-scaled RMSE, $\mathrm{VRMSE}(\hat{x},x) = \sqrt{\|\hat{x}-x\|_2^2 \,/\, (\|x-\bar{x}\|_2^2+\epsilon)}$, with $\mathrm{VRMSE}=1$ equivalent to predicting the spatial mean $\bar{x}$ [Ohana et al., 2024].

Hypotheses. **H1** (analog simulation): a quantum bottleneck with input-conditioned Rydberg dynamics improves rollout accuracy over parameter- and tuning-matched classical controls. **H2** (foundation models): the same block, as an adapter in a frozen video world model, improves physical-consistency scores over matched LoRA/MLP adapters. Each hypothesis carries a kill rule: if the matched control is statistically indistinguishable from or better than the quantum variant across seeds, the prong is reported dead in this regime.

Primary goal. One-step test VRMSE < 0.1956 on `turbulent_radiative_layer_2D`, significant under a paired per-sample Wilcoxon test over the $\sim$870 official test windows against the official CNextU-net checkpoint, across 3 training seeds (Holm-corrected).

On WaveLiT. Among compact architectures evaluated on TRL2D, WaveLiT [Sankaran et al., 2026] reports one-step VRMSE 0.1308 (30.5M) and 0.1346 (9M) using the benchmark's 4-step history and stating consistency with the official evaluation. The aggregation statistic, test split, and normalization are, however, undocumented and no code is released; the authors' extended version switches to median VRMSE (0.1167 for the same 9M model), a statistic not comparable to the benchmark's mean convention, and the 30.5M model's training budget exceeds the benchmark's 12h single-GPU convention by roughly 2–3×. We treat WaveLiT as a promising but not protocol-verified comparison point and evaluate exclusively under the official mean-VRMSE protocol on the official test split.

3 Method (summary)

A classical trunk produces a latent field z; an attention pool yields $h \in \mathbb{R}^d$; a linear map sets per-qubit, per-layer detunings $\delta = \pi\tanh(Wh)$; each 10-qubit register evolves under $H_{\mathrm{Ryd}}/\hbar = \sum_i \frac{\Omega_\ell}{2}X_i - \sum_i \delta_{\ell,i}n_i + \sum_{i<j} C_6|r_i-r_j|^{-6}n_in_j$ with trainable times τ_ℓ, global drives Ω_ℓ, and trainable coordinates r_i (geometry-as-parameter); $\langle Z_i\rangle$ readouts re-enter the trunk through a zero-initialized gated residual. The quantum core is $\sim$500–2,100 parameters per register; every baseline implements the identical pool→core→residual interface at matched total parameters ($\pm0.3\%$). Training uses a batched complex64 statevector propagator with exact autograd gradients, cross-checked against an independent simulator to 10^{-6}. Prong H2 taps the denoised chunk latent of the frozen world model at its KV-cache chunk boundary, one adapter call per chunk.

4 Pilot evidence (time-boxed sprints; internal protocol)

Pilot sprints use a stratified subset (3 of 9 cooling regimes, $2\times$ downsampled, single-frame input, 5-minute wall-clock or fixed-step budgets, $\sim$1.33M parameters, fixed seeds) — internally comparable, *not* comparable to the published table; the full-protocol runs target Section 2's goal directly.

Variant	VRMSE@1	VRMSE@8	steps	note
persistence floor	0.583	1.002	—	trivial baseline
FNO trunk (none)	0.419	1.143	10,597	
+ matched MLP control	0.425	1.213	10,326	control *hurts*
+ 12q HEA bottleneck	**0.404**	**0.876**	6,912	-23% rollout
+ 16q Rydberg-analog	0.432	0.913	2,850	beats floor
fixed 2,500 steps: none	0.477	2.102	2,500	
fixed 2,500 steps: +12q	0.486	**1.325**	2,500	-37% per-step
3-seed study (mean±std over seeds):				
none	—	1.111 ± 0.174	$\sim$10.3k	
+ matched MLP (30-dim)	—	1.123 ± 0.211	$\sim$10.1k	control
+ 30q HEA (3×10q)	—	1.063 ± 0.152	$\sim$4.4k	
+ 30q Rydberg (3×10q)	—	$\mathbf{0.955 \pm 0.102}$	$\sim$5.4k	best mean & lowest var

Table 1: Pilot sprints (The Well TRL2D subset). Only quantum variants beat the persistence floor at 8-step rollout; the effect persists at fixed step budget (throughput removed). Multi-seed replication is in progress: run-to-run $\sigma(\mathrm{VRMSE@8}) \approx 0.2$ rivals single-run effect sizes, so per-regime paired statistics (9 points/comparison) gate any claim.

Across three fixed-seed replicates, the variant ordering matches the hypothesis pattern — analog $>$ digital $>$ no bottleneck $>$ classical control — with the 30-simulated-qubit Rydberg block best on both mean and variance, and the analog qubit-scaling trend monotone in our runs (12q: 1.17 $\rightarrow$ 16q: 0.91 $\rightarrow$ 3×10q: 0.955 ± 0.102). With n=3 aggregate seeds these differences are directional, not significant; the per-regime paired protocol and the full-protocol runs below carry the statistical burden. As the anchor for the primary goal, we re-evaluated the official CNextU-net checkpoint on the official test split with our pipeline: mean VRMSE **0.1934** over 873 windows (published: 0.1956; 1.1% discrepancy), and we adopt our stricter reproduced number as the bar.

5 Results

Four result blocks, in decreasing order of maturity: a completed three-seed internal study whose numbers are final (§5.1); the official-benchmark campaign, in flight at submission-scaffolding time, whose cells are visibly marked placeholders (§5.2); released artifacts (§5.3); and the full evidence ladder with honest per-benchmark status (§5.4).

5.1 Sprint arena, wave 1.5: rollout-aware training unlocks the quantum edge

The decisive internal experiment crosses the training objective with the bottleneck variant at a fixed 2,000-step budget (throughput removed), 3 seeds: one-step MSE (k=1) versus a k=4 pushforward/TD objective (inner rollout steps stop-gradiented, loss on the final step). Results in Table 2 are complete and final.

This is the pre-registered H1a pattern (quantum $<$ both controls) at its strongest to date, and it explains the flat one-step fine-tunes observed in early checkpoint-surgery pilots: the quantum bottleneck's contribution lives in rollout dynamics, which the one-step loss never sees.

5.2 Official-benchmark campaign (TRL2D): checkpoint surgery in two rounds

The campaign fine-tunes the *official released* CNextU-net checkpoint under the official protocol, three arms (vanilla continued / matched-MLP control / 30q Rydberg bottleneck) $\times$ three seeds, with single-shot test evaluation per best-on-validation model and paired per-window Wilcoxon against our reproduced anchor (0.1934 over 873 windows; published: 0.1956). Round 1 trains with the

Variant	VRMSE@8, k=1	VRMSE@8, k=4 (TD)	TD gain
FNO trunk (none)	0.964 ± 0.005	0.957 ± 0.255	$\approx 0\%$
+ matched MLP (control)	1.057 ± 0.057	0.953 ± 0.135	-10%
+ 30q Rydberg (3×10q)	1.100 ± 0.194	$\mathbf{0.845 \pm 0.051}$	$\mathbf{-23\%}$
+ 30q Rydberg, conditioned geometry	1.043 ± 0.210	0.895 ± 0.137	-14%

Table 2: Wave 1.5 (The Well TRL2D subset, internal protocol): 8-step rollout VRMSE, mean±std over seeds {1337, 1338, 1339}, fixed 2,000 steps. Under the TD objective the 30-simulated-qubit Rydberg bottleneck beats the parameter-matched MLP control on **3/3 seeds** (0.903/0.826/0.806 vs 1.098/0.933/0.829) and posts the best mean and lowest variance overall. Interaction effect: switching to the TD objective improves the quantum variant by -23%, the classical control by -10%, and the bare trunk not at all — under one-step training the trunk alone is best and the quantum edge is *invisible*. Adapters need rollout-aware training; this finding sets the round-2 design in §5.2.

benchmark's one-step objective; round 2 repeats the surgery with the wave-1.5-informed TD objective (k=4). A 3k-step pilot preserved the hypothesis ordering on validation VRMSE (quantum 0.18977 < vanilla 0.18985 < MLP 0.19009), differences within noise — consistent with wave 1.5's finding that one-step training barely moves any arm. Numbers below are placeholders until the campaign's paired statistics land; the verdict is reported either way. Under the one-step objective, all arms improve minutely and significantly on the released checkpoint but indistinguishably (kill rule fired). Under the TD objective, evaluated on the official test rollout, the quantum arm is the only configuration that improves the long-horizon window on every seed: 13:30 VRMSE 0.9528 ± 0.0432 vs vanilla 1.0301 ± 0.0367 and matched-MLP 1.0322 ± 0.0212 (anchor 1.0132, published 1.01), winning the seed-paired comparison 3/3 against both controls while matching them on the medium window. TD arms are rollout models: their one-step error rises by design. Our dequantization suite then shows the trained adapter is *inference-inert*: product-state and χ=1 surrogates, 128-shot readout, and even zeroing the readout reproduce the fine-tuned model's rollout to $\leq 2\times10^{-5}$ (600× below seed noise; register entanglement 0.03–0.09 bits). The quantum advantage is therefore a *training-dynamics* effect — the bottleneck steers TD fine-tuning into a better basin that is then encoded in the trunk — consistent with our pre-registered framing of quantum-structured inductive bias rather than inference-time quantum computation. The pre-registered training-side ablation (2026-08-23) is decisive: retraining the identical adapter with blockade interactions zeroed — product state from step 0, same TD protocol, three seeds — loses the effect entirely, giving 13:30 test VRMSE 1.0156 ± 0.0433 with only 1/3 seeds below the anchor (vs 3/3 entangled), while the entangled arm's rollout-validation selection is better on every seed. Entanglement is thus dispensable at inference yet necessary during optimization: the gain is specific to training *through* entangled Rydberg dynamics, not to a generic high-dimensional bottleneck.

Round	Arm	Test VRMSE (3 seeds)	Wilcoxon p vs anchor	win rate
—	official checkpoint (anchor)	0.1934	—	—
1 (one-step)	vanilla continued	0.19307 ± 0.00007	—	—
1 (one-step)	+ matched MLP (control)	0.19314 ± 0.00016	—	—
1 (one-step)	+ 30q Rydberg	0.19315 ± 0.00007	$\leq 1.9\times10^{-2}$ per seed (best 6×10^{-12})	52.5–60.1%
2 (TD, k=4)	vanilla continued	1.0301 ± 0.0367	—	—
2 (TD, k=4)	+ matched MLP (control)	1.0322 ± 0.0212	—	—
2 (TD, k=4)	+ 30q Rydberg	$\mathbf{0.9528 \pm 0.0432}$	3/3 seeds paired vs both controls	3/3 below anchor

Table 3: TRL2D official-protocol campaign (scaffolding; red cells are in-flight placeholders). One-step test mean-VRMSE on the official test split; paired one-sided Wilcoxon (arm < anchor) over 873 per-window scores, Holm-corrected across seeds. Pass rule (pre-registered H1d): quantum below both controls *and* below the 0.1934 anchor at $p < 0.05$. Kill rule: if the matched control is indistinguishable from or better than the quantum arm, the prong is reported dead in this regime and this table ships with that verdict.

5.3 Released artifacts

TORAX-Bench v1. With this paper we release a new benchmark: 1,000 actuator-conditioned tokamak core-transport discharges generated with the TORAX simulator (all four transport equations evolved — ion and electron heat, current, electron density), sampled i.i.d. from a 12-dimensional scenario space with *time-varying* actuators (plasma-current ramp, three-knot heating waveform, gas-puff step). Each discharge saves $[T_e, T_i, n_e, q]$ profiles on a uniform 121-frame, 6-second grid plus the commanded actuator waveforms; splits are frozen at 800/100/100 with a dataset card and full generation manifest (44 MB total). Reference CPU baselines are set and frozen: persistence@8 mean rollout VRMSE **0.0923** (the number to beat) and per-timestep climatology 0.5546. Adapter numbers on this benchmark are pending and not claimed here.

Pre-registered protocol and reproduction pack. The second artifact is the protocol itself: the hypothesis scoreboard with explicit pass/kill rules per claim, the per-run results ledger, the paired-statistics scripts, and a `REPRODUCE.md` that maps every number appearing in this paper — including every pending cell above — to the exact command or executed notebook that regenerates it. All runs execute on a single 126 GB workstation.

5.4 Evidence ladder

Table 4 is the full claim surface with per-benchmark status at scaffolding time — filled, in-flight, and not-yet-started rows alike. Rows convert from "pending" to numbers only through the pre-registered commands; nothing is claimed for a pending row.

Benchmark	Metric	Reference point	Ours	Status
The Well TRL2D (official)	VRMSE ↓	CNextU-net 0.1956 / 0.1934	Table 3	campaign in flight
TRL2D sprint arena (internal)	VRMSE@8 ↓	best classical 0.953 ± 0.135	$\mathbf{0.845 \pm 0.051}$, 3/3 seeds	**complete**
The Well MHD_64	VRMSE ↓	CNextU-net 0.1633	—	pipeline ready
TORAX-Bench v1 (ours)	VRMSE@8 ↓	persistence@8 0.0923	baselines released	adapter pending
Physics-IQ	score ↑	Cosmos ≈ 63.4	—	pipeline staged
WorldModelBench	judge score ↑	leaderboard	—	queued (full version)
TokaMark G2 (MAST magnetics)	NRMSE ↓	CNN+LSTM 0.1343	—	data access verified
Sophelio equilibrium challenge	G_{ratio} ↑	Ch2 open ≈ 0.654	—	committed (Oct 2026)
Pelzer 1HP heat plume	MAE ↓	U-Net 0.046 °C	—	loader + split frozen

Table 4: Evidence ladder: one adapter stack, every benchmark it is pre-registered against, and the honest status of each row. "Reference point" is the published or strongest available number for that benchmark; dashes are not results. The two filled rows are the sprint-arena study (Table 2) and the TORAX-Bench baseline release (§5.3).

6 Roadmap and limitations

Full-protocol runs fine-tune the official CNextU-net checkpoint with and without the quantum bottleneck under identical budgets. Known limitations we carry openly: the pooled register is a poly(n) information bottleneck (readout scaling is reported, not hidden); entangled 30-qubit training uses bounded-χ tensor-network simulation (exact statevector at 30q is inference-only on our hardware); sprint-scale variance mandates the paired protocol above; and all quantum components are classically simulated, so hardware execution (via Pulser export to neutral-atom platforms) is future work, not a claim. All experiments, including every negative result, run on a single 126 GB workstation and reproduce from one command per table.

References

Vytautas Abramavicius, Melvin Mathé, Gergana V. Velikova, João P. Moutinho, Mario Dagrada, Vincent E. Elfving, Alexandre Dauphin, Joseph Vovrosh, and Roland Guichard. PulserDiff: A pulse differentiable extension for Pulser. *arXiv preprint arXiv:2505.16744*, 2025.

Borja Aizpurua, Sukhbinder Singh, Augustine Kshetrimayum, Saeed S. Jahromi, and Roman Orus. Quantum-enhanced large language models on quantum hardware via cayley unitary adapters. *arXiv preprint arXiv:2605.05914*, 2026.

Daniel Barredo, Vincent Lienhard, Sylvain de Léséleuc, Thierry Lahaye, and Antoine Browaeys. Synthetic three-dimensional atomic structures assembled atom by atom. *Nature*, 561:79–82, 2018. Optical-tweezer assembly of atoms into arbitrary user-defined 3D geometries.

Joseph Bowles, Shahnawaz Ahmed, and Maria Schuld. Better than classical? The subtle art of benchmarking quantum machine learning models. *arXiv preprint arXiv:2403.07059*, 2024.

M. Cerezo, Martin Larocca, Diego García-Martín, N. L. Diaz, Paolo Braccia, Enrico Fontana, Manuel S. Rudolph, Pablo Bermejo, Aroosa Ijaz, Supanut Thanasilp, Eric R. Anschuetz, and Zoë Holmes. Does provable absence of barren plateaus imply classical simulability? *Nature Communications*, 16:7907, 2025. arXiv:2312.09121.

Nicholas S. DiBrita, Daniel Leeds, Yuqian Huo, Jason Ludmir, and Tirthak Patel. ReCon: Reconfiguring analog rydberg atom quantum computers for quantum generative adversarial networks. In *Proceedings of the International Conference on Computer-Aided Design (ICCAD)*, 2024. arXiv:2408.13389.

Nicholas S. DiBrita, Jason Han, and Tirthak Patel. ResQ: A novel framework to implement residual neural networks on analog rydberg atom quantum computers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2506.21537.

Zelin Gao, Qiuyu Wang, Jiapeng Zhu, Jingye Chen, Zichen Liu, Qingyan Bai, Jiahao Wang, Yufeng Yuan, Hanlin Wang, Yichong Lu, Ka Leong Cheng, Haojie Zhang, Jian Gao, Tianrui Feng, Yuzheng Liu, Yao Yao, Yinghao Xu, Xing Zhu, Yujun Shen, and Hao Ouyang. Infinite worlds with versatile interactions. *arXiv preprint arXiv:2607.07534*, 2026. LingBot-World 2.0 (Infinity) technical report, Robbyant / Ant Group. Models: `https://huggingface.co/robbyant/lingbot-world-v2-14b-causal-fast`.

Md Aminur Hossain, Ayush V. Patel, Siddhant Gole, Sanjay K. Singh, and Biplab Banerjee. HQF-Net: A hybrid quantum-classical multi-scale fusion network for remote sensing image segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026. arXiv:2604.06715.

He-Liang Huang, Yuxuan Du, Ming Gong, Youwei Zhao, Yulin Wu, Chaoyue Wang, Shaowei Li, Futian Liang, Jin Lin, Yu Xu, Rui Yang, Tongliang Liu, Min-Hsiu Hsieh, Hui Deng, Hao Rong, Cheng-Zhi Peng, Chao-Yang Lu, Yu-Ao Chen, Dacheng Tao, Xiaobo Zhu, and Jian-Wei Pan. Experimental quantum generative adversarial networks for image generation. *Physical Review Applied*, 16:024051, 2021. doi: 10.1103/PhysRevApplied.16.024051. arXiv:2010.06201; patch-method quantum GAN.

Naman Jain and Amir Kalev. QuFeX: Quantum feature extraction module for hybrid quantum-classical deep neural networks. *Quantum Science and Technology*, 11:015017, 2026. arXiv:2501.13165.

Hailong Jiang, Emran Hossain, Feng Yu, Jianfeng Zhu, Guilin Zhang, and Wulan Guo. Quantum-structured world models (QSWMs) for predictive latent dynamics. *arXiv preprint arXiv:2608.05371*, 2026. Quantum-inspired: complex-valued, density-matrix-like latents evaluated on cellular automata; no quantum module.

Sang Hyub Kim, Jonathan Mei, Claudio Girotto, Masako Yamada, and Martin Roetteler. Quantum large language model fine-tuning. *arXiv preprint arXiv:2504.08732*, 2025. IonQ; accuracy scaling with qubit count against matched classical baselines.

Toshiaki Koike-Akino, Francesco Tonin, Yongtao Wu, Frank Zhengqing Wu, Leyla Naz Candogan, and Volkan Cevher. Quantum-peft: Ultra parameter-efficient fine-tuning. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2503.05431.

Milan Kornjača, Hong-Ye Hu, Chen Zhao, Jonathan Wurtz, Phillip Weinberg, Majd Hamdan, et al. Large-scale quantum reservoir learning with an analog quantum computer. *arXiv preprint arXiv:2407.02553*, 2024. Experiments on up to 108 qubits on a neutral-atom analog quantum computer (QuEra Aquila).

Dacheng Li, Yunhao Fang, Yukang Chen, Shuo Yang, Shiyi Cao, Justin Wong, Michael Luo, Xiaolong Wang, Hongxu Yin, Joseph E. Gonzalez, Ion Stoica, Song Han, and Yao Lu. Worldmodelbench: Judging video generation models as world models. *arXiv preprint arXiv:2502.20694*, 2025.

Hannah J. Manetsch, Gyohei Nomura, Elie Bataille, Kon H. Leung, Xudong Lv, and Manuel Endres. A tweezer array with 6100 highly coherent atomic qubits. *Nature*, 647:60–67, 2025.

Paolo Marcandelli, Yuanchun He, Stefano Mariani, Martina Siena, and Stefano Markidis. Partitioned hybrid quantum fourier neural operators for scientific quantum machine learning. *arXiv preprint arXiv:2507.08746*, 2025.

Michael McCabe, Payel Mukhopadhyay, Tanya Marwah, Bruno Regaldo-Saint Blancard, Francois Rozet, Cristiana Diaconu, Lucas Meyer, Kaze W. K. Wong, Hadi Sotoudeh, Alberto Bietti, Irina Espejo, Rio Fear, Siavash Golkar, Tom Hehir, Keiya Hirashima, Geraud Krawezik, Francois Lanusse, Rudy Morel, Ruben Ohana, Liam Parker, Mariel Pettee, Jeff Shen, Kyunghyun Cho, Miles Cranmer, and Shirley Ho. Walrus: A cross-domain foundation model for continuum dynamics. *arXiv preprint arXiv:2511.15684*, 2025.

Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. Do generative video models understand physical principles? *arXiv preprint arXiv:2501.09038*, 2025. Physics-IQ benchmark, 66 real-world scenarios. Code: `https://github.com/google-deepmind/physics-IQ-benchmark`.

Ruben Ohana, Michael McCabe, Lucas Meyer, Rudy Morel, Fruzsina J. Agocs, Miguel Beneitez, Marsha Berger, Blakesley Burkhart, Keaton Burns, Stuart B. Dalziel, Drummond B. Fielding, Daniel Fortunato, Jared A. Goldberg, Keiya Hirashima, Yan-Fei Jiang, Rich R. Kerswell, Suryanarayana Maddu, Jonah Miller, Payel Mukhopadhyay, Stefan S. Nixon, Jeff Shen, Romain Watteaux, Bruno Régaldo-Saint Blancard, François Rozet, Liam H. Parker, Miles Cranmer, and Shirley Ho. The well: A large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Datasets and Benchmarks Track*, 2024. arXiv:2412.00568.

Tushar Pandey. When classical baselines are tuned as carefully as the quantum model, does quantum reservoir computing still win? *arXiv preprint arXiv:2607.09905*, 2026.

Mateusz Papierz, Asel Sagingalieva, Alix Benoit, Toni Ivas, Elia Iseli, and Alexey Melnikov. Hybrid fourier neural operator for surrogate modeling of laser processing with a quantum-circuit mixer. *arXiv preprint arXiv:2604.04828*, 2026. Introduces HQ-LP-FNO.

Snehal Raj and Brian Coyle. Quic: Quantum-inspired compound adapters for parameter efficient fine-tuning. *arXiv preprint arXiv:2502.06916*, 2025.

Shyam Sankaran, Hanwen Wang, and Paris Perdikaris. Small models, strong priors: Architectural inductive bias for parameter-efficient neural pde solvers. *arXiv preprint arXiv:2605.25949*, 2026. Introduces WaveLiT (1–10M-parameter wavelet linear-attention transformer).

Maria Schuld and Nathan Killoran. Is quantum advantage the right goal for quantum machine learning? *PRX Quantum*, 3:030101, 2022. doi: 10.1103/PRXQuantum.3.030101. arXiv:2203.01340.

Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.

Wei Xia, Shuaifan Cao, Xingze Qiu, and Xiaopeng Li. Quantum magic and non-commutativity as computational resources in quantum reservoir computing. *arXiv preprint arXiv:2607.12035*, 2026. Shows Hamiltonian input encoding yields transcendental, infinite-order nonlinearity in the reservoir update map.

Yumiao Zhao, Bo Jiang, Min Lu, Xiao Wang, and Jin Tang. Mqadapter: Multi-modal quantum adapter for coarse-to-fine vlm fine-tuning. *arXiv preprint arXiv:2607.12418*, 2026.